europass



Most Rimi Moni

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ABOUT ME

Professional Profile

Highly motivated Computer Science and Engineering graduate (CGPA: 3.73/4.00) with proven undergraduate research experience in Machine Learning, Explainable AI (XAI), and Cyber Security. Passionate about developing lightweight offline intelligent systems, data-driven pedagogical analytics, and trustworthy AI. Seeking to leverage strong technical expertise and interdisciplinary research capabilities in an advanced academic degree program.

EDUCATION & TRAINING

Bachelor of Science in Computer Science and Engineering

Canadian University of Bangladesh (CUB) [20/02/2022 - 23/10/2025]

Address: 201/1 Pragati Sharani (Bir Uttam Rafiqul Islam Avenue).1212, Dhaka (Bangladesh) | **Website:** https://cub.edu.bd/ | **Field(s) of study:** Computer Science and Engineering: | **Final grade:** 3.73 out of 4 | **Level in EQF:** 6 | **Number of credits:** 150 | **Thesis:** Phishing URL Detection Using Machine Learning: A Lightweight Offline Approach

Adviser: Md. Hasanuzzaman (Head of the Department)

Focus: Phishing URL Detection, Machine Learning, Lightweight Offline Systems, Zero-Network-Dependency (ZND), Lexical Feature Engineering.

Key Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Cyber Security.

RESEARCH EXPERIENCE

Undergraduate Research Work

[08/2025 – 12/2025]

Project: A Hybrid Stacking Ensemble Framework for Predicting Mathematics Avoidance Behavior: Integrating Explainable AI (XAI) for Pedagogical Sustainability in Bangladesh.

Research Supervisors: Tahmina Islam, Tajel Hossan

Focus: Mathematics Anxiety, Pedagogy, Explainable AI (XAI), Machine Learning

Key Contribution: Conducted a thorough literature review to identify key behavioral indicators and establish a theoretical framework for analyzing Bangladeshi student performance data. Curated, cleaned, and sorted an original dataset of 1,953 records, transforming raw survey responses into a standardized structure optimized for predictive modeling. Applied exploratory data analysis (EDA) and built a hybrid stacking ensemble model to uncover behavioral patterns and enhance performance forecasting.

Published Dataset: https://data.mendeley.com/datasets/mypg6544zr/1

PUBLICATIONS

[2026]

SMADS: Student Mathematics Anxiety and Digital Learning Modalities Survey Dataset for Predictive Modeling

Authors: Most Rimi Moni, et al. | **Repository:** Mendeley Data (Elsevier) | **DOI:** 10.17632/mypg6544zr.1

Published an open-access, preprocessed cross-disciplinary dataset ($N = 1,953$, 16 variables) capturing student demographics, study habits, mathematics anxiety, pre-test stress, teacher adaptability, and exam outcomes across Traditional, AI-Based, and Hybrid learning modalities in Bangladesh.

Link: https://doi.org/10.17632/mypg6544zr.1

PROJECTS

Advanced Predictive Modeling Portfolio Developed a suite of Machine Learning models to solve diverse classification and regression problems.

- **Credit Scoring Model:** Engineered a supervised machine learning pipeline in Python to assess consumer creditworthiness and financial default risk; optimized precision-recall metrics using custom feature engineering on institutional financial records.
- **Clinical Diagnostic Classifiers:** Architected diagnostic classification models utilizing Support Vector Machines (SVM) and ensemble techniques (PIMA dataset) to forecast disease susceptibility with high precision.
- **Automated Loan Approval Engine:** Developed an end-to-end decision pipeline automating credit evaluations based on demographic and historical financial variables.
- **Tools used:** Python, Scikit-learn, Pandas, Matplotlib, Seaborn.

Link: https://github.com/riminipa16

Computer Vision & Specialized Analytics Applied deep learning and signal processing techniques to complex datasets.

- **Handwritten Character Recognition:** Designed and trained a deep convolutional neural network (CNN) to detect, segment, and classify handwritten alphanumeric characters.
- **SONAR Underwater Object Detection:** Built a high-accuracy binary classifier using Logistic Regression to analyze acoustic sonar signals for underwater obstacle/mine identification.
- **Stock Market Volatility Analysis:** Formulated time-series statistical indicators and rolling-window algorithms to quantify market volatility and cross-asset correlations.
- **Tools used:** Jupyter Notebook, XGBoost, NumPy, Pandas.

Link: https://github.com/riminipa16

SKILLS

Programming Languages

SQL | C | C++ | Java (computer programming) | Python

Web Development

CSS & HTML | JavaScript | WordPress

Data Science & ML Libraries

XGBoost | scikit-Learn | Seaborn | Matplotlib | Numpy | Pandas | SHAP

Software & Tools

Git | Jupyter NoteBook | Adobe Creative Suite

WORK EXPERIENCE

Fiverr - Bangladesh, Online

Graphic designer

[01/06/2020 - Current]

Achieved and maintained **Level 1 Rank** seller status through high-quality delivery and customer satisfaction.

Self-Employed - Bangladesh, Dhaka

Private tutor

[01/05/2022 - Current]

Actively teaching English medium students for over 4 years to enhance their academic performance.

CERTIFICATIONS

[Saylor University, 20/10/2026]

CS250: Python for Data Science

[Saylor University, 11/11/2026]

CS207: Fundamentals of Machine Learning

HONOURS AND AWARDS

[11/04/2022] Canadian University of Bangladesh

Kickoff Programming Contest CUB I secured 8th place in the kickoff Programming contest held in Canadian University of Bangladesh.

[28/09/2022] Canadian University of Bangladesh

Course Project Award CUB 1st Runner-Up Position

[01/06/2016] Bangladesh Math Olympiad

1st runners up in Math Olympiad (Regional) In 2016 regional math olympiad contest I secured the first runners up place from my school.

VOLUNTEERING

Dinajpur

LIBMUN 2018 Supported in LIBMUN-18 as a Director of Conference Management

Dinajpur

Dinajpur Flash Flood 2017 With the Help of Dinajpur School of Liberators me and my friends distributed releifs to the affected people of that flood.

LANGUAGE SKILLS

Mother tongue(s): Bengali

Other language(s): English